Literature-Grounded Novelty Assessment of Scientific Ideas

Simra Shahid[1] **Marissa Radensky**[2] **Raymond Fok**[2]
Pao Siangliulue[3] **Daniel S. Weld**[3] **Tom Hope**[3]
[1]Microsoft [2]University of Washington [3]Allen Institute for AI
simrashahid@microsoft.com {radensky, rayfok}@cs.washington.edu
{paos, danw, tomh}@allenai.org

Abstract

Automated scientific idea generation systems have made remarkable progress, yet the automatic evaluation of idea novelty remains a critical and underexplored challenge. Manual evaluation of novelty through literature review is labor-intensive, prone to error due to subjectivity, and impractical at scale. To address these issues, we propose the **Idea Novelty Checker**, an LLM-based retrieval-augmented generation (RAG) framework that leverages a two-stage retrieve-then-rerank approach. The Idea Novelty Checkerfirst collects a broad set of relevant papers using keyword and snippet-based retrieval, then refines this collection through embedding-based filtering followed by facet-based LLM re-ranking. It incorporates expert-labeled examples to guide the system in comparing papers for novelty evaluation and in generating literature-grounded reasoning. Our extensive experiments demonstrate that our novelty checker achieves approximately 13% higher agreement than existing approaches. Ablation studies further showcases the importance of the facet-based re-ranker in identifying the most relevant literature for novelty evaluation.

1 Introduction

Novelty evaluation is foundational for determining whether ideas in scientific research, product development, or creative ideation introduce meaningful innovation relative to prior work. Yet, as the volume of published literature grows exponentially, manual verification of originality becomes impractical. This is further complicated by the inherent subjectivity of novelty judgments, which is why experts can more easily decide on similarity of two ideas (Picard et al.) and often struggle to articulate why one idea is more novel than another. Further the evaluation becomes subjective as it also depends on personal knowledge and intuition gained from scientific literature (Ahmed et al.; Picard et al.).

Automated systems attempt to address this challenge by defining novelty as differences observed while comparing new ideas against prior work with similarity measures, but they exhibit critical limitations. Prior work has evolved from using n-gram frequency and lexical metrics (TF-IDF, LDA) (Wang et al., b; Sarica et al.) to semantic embeddings (Gómez-Pérez et al.; Su et al.) that capture similarity but don't capture paraphrased variations of ideas and papers. Moreover, while recent approaches have adopted LLM-augmented pipelines to generate numerical scores (1-10) (Bougie and Watanabe; Wang et al., c) or provide binary classifications (novel versus not novel) (Lu et al.; Li et al.; Si et al.; Su et al.), they do not ground the rationales in existing works and frequently fail to capture subtle variations in phrasing, resulting in the misclassification of well-documented ideas as novel (Beel et al.; Gupta and Pruthi). This shortcoming makes it difficult for researchers to distinguish novel ideas from incremental contributions or subtle cases of plagiarism (Gupta and Pruthi).

Moreover, all these approaches hinge on the successful retrieval of relevant literature for a given idea, a task that remains inherently challenging (Mysore et al., a,b; Stevenson and Merlo; Eger et al.; Xu et al.; Mahajan et al.). Prior works (Si et al.; Lu et al.) extract keywords from the idea to search for papers, so important work can easily be missed if the relevant paper does not have the exact keyword. This undermines the reliability of the novelty evaluation process.

We address these gaps with Idea Novelty Checker, a retrieval-augmented LLM pipeline that assesses an idea's novelty by comparing it to a set of the most relevant papers. First, Idea Novelty Checkercollects a broad set of relevant papers using keyword and snippet-based retrieval, as well as by retrieving papers similar to any seed papers provided. Next, an embedding-based similarity search filters this large collection, and a facet-based LLM re-ranker (Sun et al.) further narrows the set by comparing idea facets (purpose, mechanism, evaluation, and application) with those in the retrieved papers. Finally, expert-annotated in-context examples of *novel* and *not novel* ideas guide the system in generating literature-grounded rationales, mitigating subjectivity in novelty judgments.

In our experiments, we compared Idea Novelty Checkerwith baselines such as zero-shot prompting, prompt optimization approaches (DSPY and TextGRAD), and expert-based OpenReview examples. Our results show that expert-annotated in-context examples significantly improve classification performance. Comparisons with systems like AI Scientist and AI Researcher further demonstrate that our Idea Novelty Checkerachieves higher agreement with expert judgments, and our ablation studies shows that the combined retrieval and two-stage re-ranking are critical for identifying the most relevant papers.

Our contributions are as follows:

- We introduce Idea Novelty Checker, a retrieval-augmented LLM pipeline that automatically evaluates the novelty of scientific ideas. We plan to release our code and expert-collected data[1] to support work in automatic scientific discovery and provide literature-grounded novelty evaluations.

- We conducted a formative study in which experts evaluated ideas for novelty. The study revealed two key challenges to consider for novelty evaluation: clarifying what constitutes novelty given its subjectivity, and identifying relevant literature to assess it. This directly shaped the design of Idea Novelty Checker.

- Our method integrates keyword-based and snippet-based retrieval, followed by a two-stage re-ranker with embedding similarity and facet-based LLM re-ranking to identify key literature related to the given idea.

- We present extensive evaluations, ablation studies, and qualitative analyses that demonstrate the effectiveness of our novelty checker over existing approaches. Additionally, we discuss prompt sensitivity in LLMs for novelty evaluation further highlighting the importance of clear novelty definitions.

[1] github.com/simra-shahid/idea_novelty_checker

2 Related Work

Automated approaches to novelty assessment in scientific literature have evolved considerably. Early methods relied on lexical similarity metrics, such as TF-IDF, LSA, and LDA (Wang et al., b; Sarica et al.), but these techniques struggled to capture paraphrased concepts. Semantic embedding methods (Gómez-Pérez et al.) improved on this by identifying deeper relationships, yet they are confined to surface-level comparisons (Mysore et al., a,b).

Retrieval-augmented LLM systems have emerged as a promising alternative, evaluating novelty either on a numerical scale (e.g., 1–10) (Bougie and Watanabe; Wang et al., c) or with binary classification (Li et al.; Lu et al.). AI Researcher (Si et al.) uses a Swiss-system tournament ranking to compare ideas pairwise for *similarity* and *novelty* against individual papers. If any comparison has sufficient similarity, the idea is not novel. Another notable work is AI Scientist (Lu et al.) that employs an iterative process in which an LLM generates queries from a research idea to retrieve relevant papers via the Semantic Scholar API (Kinney et al.). The LLM then compares the idea against these papers until a clear decision is reached or a preset iteration limit is met. However, this approach has several limitations. First, it depends on keyword-based retrieval methods to get the most relevant papers to an idea, which may fail if the relevant papers do not contain the exact keywords. Second, comparing an idea against a large number of retrieved papers (sometimes over 100) can introduce known issues that LLMs often overlook instructions within a prompt (Loya et al.; Sclar et al.; Joshi et al.). Finally, the decision of novelty evaluation relies on string matching for phrases like "decision made: novel" or "decision made: not novel." If such a decision is not reached, the idea is automatically considered novel. Independent evaluations (Beel et al.) have further highlighted challenges in AI Scientist's novelty assessments, noting that the system can misclassify well-established concepts (micro-batching for stochastic gradient descent) as novel.

Our work builds on these insights by combining retrieval-then-rerank methods (Zhou et al.; Naik et al.) and uses expert-annotated examples to ensure that our novelty evaluations are grounded in the relevant literature.

3 Formative Study on Challenges in Evaluating Novelty

Evaluating idea novelty in scientific literature is inherently challenging because the criteria for novelty are subjective and can be defined in multiple ways. We conducted a formative study, referred to as the expert-annotated study throughout the paper, where the first and second authors reviewed the novelty of ideas based on the most relevant papers.

To assess idea novelty relative to existing literature, our study engaged experts who evaluated 51 ideas, comprising of 46 generated by the Scideator system (Radensky et al.) and 5 adapted from accepted and rejected papers from OpenReview (ICLR 22, NeurIPS 23).[2] Each idea was classified into one of three categories: novel, moderately novel, or not novel. For every idea, we identified the most relevant papers through a two-step process: candidate papers were initially gathered using keyword-based queries and subsequently re-ranked using an LLM-based reranker (Sun et al.) according to their overall relevance to the idea.

The experts achieved a moderate agreement (Cohen's Kappa = 0.64). A key challenge identified was that experts sometimes relied on their broader domain knowledge rather than restricting their judgments to the most relevant papers, as the top papers alone were often not sufficient. Additionally, using three categories led to disagreements, as the distinction between novel and moderate novelty is itself subjective.

Building on these observations, we conducted a second study to minimize the influence of external knowledge. In this study, experts were instructed to base their judgments solely on the provided papers, and the categories were simplified to just two: novel and not novel.

Inspired by prior work (Portenoy et al.; Kang et al., b; Chan and Chang; Suh et al.; Srinivasan and Chan; Choi et al.; Kang et al., a; Radensky et al.) that categorizes research ideas into core facets such as purpose (the problem being addressed by the paper) and mechanism (the proposed solution to the problem), we define novelty as follows: An idea is considered novel if it differs from all retrieved papers in at least one core facet for the topic at hand—namely, purpose (i.e., a distinct objective), mechanism (i.e., a distinct technical approach), or evaluation (i.e., a distinct validation method). An idea is also considered novel if it uniquely combines these facets or applies them to a new application domain.

Using this controlled framework, we reannotated a set of ideas and evaluated 51 ideas, comprising of 34 new ones generated by Radensky et al. and 17 from the previous study where external knowledge had influenced novelty judgments. By narrowing the focus to the relevant papers alone, we observed fewer disagreements and achieved a higher agreement rate (Cohen's Kappa = 0.68). Of the 8 instances of disagreement, in 4 cases one expert overlooked details from the paper, in 2 cases the experts differed in their perception of subtle contributions to novelty, and in the remaining 2 cases no specific comments were provided.

This formative study highlights that a robust novelty checker depends critically on high-quality retrieval and a well-defined notion of novelty. These findings directly inform our methodology described in the following section.

4 Methodology: Idea Novelty Checker

Based on our formative study findings, our novelty checker is designed with two key components that address the two critical challenges: C1 ensuring high-quality retrieval of papers relevant to the idea for novelty assessment and C2 establishing clear criteria for judging novelty. The challenge C1 arises from the vast space of overlapping papers—there are hundreds of millions of potential matches. To address this, our system first filters the scientific literature to collect the most relevant papers for a given idea (see Section 4.1 and Step 1 and 2 in Figure 1). The input idea is then compared to each paper in this collection by prompting an LLM (see Section 4.2).

In tackling challenge C2, which arises from the inherent subjectivity and multiple definitions of novelty, the novelty checker leverages expert-labeled examples of *novel* and *not novel* ideas from the formative study. It generates reasoning grounded not only in comparisons against the most relevant papers but also in the standardized definition of novelty introduced earlier, which helps counteract subjectivity (see Step 3 in Figure 1). Below, we detail these two components.

4.1 Most Relevant Papers to Idea

Following established information retrieval practices (Gao et al.; Abdallah et al.; Nouriinanloo and

[2] Fewer examples were taken from OpenReview since the primary focus was on evaluating ideas from Scideator.